

not necessary that all drones require AI technology. But almost every company that deals with autonomous unmanned aerial vehicles (UAVs) or drones claims the use of AI, machine learning or deep learning. Drones often generate large amounts of data. The faster and more accurate the data that drones generate, the faster the images can be evaluated by the system and the better for the drone to fulfill the mission or meet the requirements.

AI technology enables machines to interact with the inbuilt system and operators in an intelligent way. Through AI and machine learning, the drone can perform object detection, image processing, face detection and recognition, counting, segmentation, tracking, thermal detection, fire detection, motion planning, and others.

Machine learning algorithms are designed in such a way that they can learn and improve over time when exposed to new data. Since many AI-related tasks for drones are dealing with image processing, the drone must be able to perceive the environment or objects in some way. This process is referred to as machine perception, which is an important part of any autonomous drone. Computer vision is concerned with the automatic extraction, analysis, and understanding of useful information from one or more images. Once the drone has captured raw sensor data, it usually needs to be analysed to extract meaningful information.

Many companies are trying to develop a drone to fly Beyond Visual Line of Sight (BVLOS) for maximum efficiency. For that, motion planning is an important part. The prerequisite for motion planning is usually machine perception that captures the environment. The drone visualises the environment, identifies its surroundings and the distance. Here, deep learning is deployed to detect and recognise objects including humans, vehicles, trees, buildings, and subsequently create a corresponding flight route.

The use of AI technology makes drones smarter and gives ground-level operators a human-like eye-in-the-sky. With AI algorithm, drones can sense their surroundings, which empowers them to map areas, track objects,



Fig. 2: Ehang hybrid drone (Credit: www.ehang.com)

and provide analytical feedback in real-time. AI technology helps drones increase efficiency and safety.

Hybrid power drones

Hybrid drones are almost similar to what we have seen in hybrid-electric vehicles (HEV) powered by internal combustion engine (ICE) systems and electric propulsion systems. In conventional drones, batteries (often lithium polymer or LiPo) are used as a power source. One of the problems with LiPo battery is that when it gets exhausted after use, it takes long time to charge up. So, you cannot put the drone into a continuous flying mode for longer operations.

One of the solutions to overcome this problem is hybrid technology. You can build a hybrid drone by adding the ICE and DC generator in the conventional drone. DC generator runs with the miniature ICE, which is used for charging the battery. ICEs are usually powered by energy-dense fuels such as gasoline or diesel, or liquids derived from fossils. In most hybrid drones, gasoline is used along with LiPo batteries.

The capabilities of any drone are defined by endurance and range. The endurance for a drone can be described as the total time for which a drone can fly non-stop. It depends on many factors including the capacity of the battery and the motor current, aircraft size, and payload weight.

The range indicates the total distance a drone can fly. The range depends on various factors such as

List of some hybrid drone manufacturers

- eHang
- Foxtech
- ITRI
- Quaternium
- Skyfront
- Top Flight Technologies
- ULC Robotics
- Volocopter
- Walkera
- Workhorse Surefly

the amount of current the aircraft is producing, battery, radio controller/transmitter range, endurance, flight speed, and aerodynamic performance. Note that some hybrid drones are controlled by radio-controlled transmitters or smartphones, while others are controlled through a computer or laptop. The range of this controlling device is also an important factor for the drone to cover a longer distance.

The advantages of both fuel power and electric power are used in a hybrid drone for increasing the endurance ten times more than a normal drone. When the battery is discharged, it gets charged through the DC generator while the drone is flying. Other benefits of hybrid power include higher payloads, more powerful, and higher reliability.

Applications of hybrid drones

Hybrid drones will find applications not only in the military but in several